

I teach Cambridge Computing, Digital Literacy and Mathematics at a school in Accra, and I build the software that school runs on — timetabling, grading, diagnostics — then keep going into the systems underneath. **I am the domain expert and the engineer**, which is why the modelling is precise: when I specify a grading rule or a bell schedule, I am describing a process I have worked inside. Several of these systems are in daily production use; the rest are research I can defend line by line.

CORE COMPETENCIES

SYSTEMS & LANGUAGES Rust, TypeScript, Python, C++, SQL, JavaScript, Google Apps Script	DESKTOP, WEB & MOBILE Tauri v2 (desktop and Android), React 18/19, Next.js, Vite, Tailwind, WebAssembly, Web Workers, Three.js
DATA, STORAGE & SYNC SQLite (rusqlite, WAL), Firebase (Firestore, Auth, Functions), IndexedDB, PostgreSQL, Supabase	ARCHITECTURE & AI Offline-first design, constraint satisfaction (MRV, backtracking), Gemini multimodal pipelines, alpha-beta search, NNUE, applied cryptography

WORK INDEX Line counts exclude dependencies

SYSTEM	DOMAIN	STACK	SCALE	STATUS
BaseCamp Diagnostics	AI learning-gap diagnostics	React 19 · Firebase	~95k LOC	PILOT
SAIS Portal	School information system	React · Firebase	5 yrs data	LIVE
EduScheduler Pro	Constraint-solving timetables	Tauri v2 · React 18	~72k LOC	V1.0.0
Nexus Inventory	Native POS & thermal printing	Rust · Tauri v2	127 tests	DELIVERED
Victor Draughts Engine	Game AI: search, NNUE	Rust · WASM	~30k LOC	JAN 2027
Doxamix Foods	E-commerce & admin	React 19 · Firebase	26 tests	LIVE
Academic Compliance	AI lesson-plan auditing	React · Gemini	10 depts	LIVE
HecTech Report Generator	AI report cards in Sheets	Apps Script	39 sheets	LIVE
QuantSports Engine	Quantitative odds research	TypeScript	Backtested	RESEARCH
3D Ludo Engine	Real-time 3D multiplayer	Three.js · ws	WebGL	PLAYABLE

Computing, Digital Literacy & Mathematics Teacher
St. Adelaide International Schools, Dansoman & Aburi campuses, Accra — August 2021 to present. Deliver the full Cambridge Computing and Digital Literacy curriculum and Cambridge Mathematics. Founded and lead the school Coding Club. Built and maintain the institutional software in daily administrative use.

Database Manager
UMB Bank, Accra — 2019 to 2020. Schema integrity, access control and routine maintenance, and the backup and recovery procedures that guaranteed data continuity. Where the habit of treating data loss as unacceptable comes from.

BSc Telecommunication Engineering
Ghana Technology University College, degree awarded by KNUST, conferred 2018. Engineering Mathematics I-V, Signals and Systems, Information Theory, Digital Signal Processing. Principles of Computer Programming (A), Algorithms and Data Structures (B).

Recognition
UNICEF StartUp Lab — Learning Gaps Challenge, Certificate of Participation, March 2026, for BaseCamp Diagnostics. Google AI Essentials, 2026. Seven British Council certifications in classroom practice, 2022–2025.

FLAGSHIP — EDUCATION TECHNOLOGY

BaseCamp Diagnostics PILOT

AI-assisted learning-gap diagnostics for classrooms with unreliable connectivity

SCALE	STACK	ROLE	LIVE
~95k lines of source	React 19 · TypeScript · Firebase · Gemini · idb- keyval	Sole architect and engineer. I teach the numeracy curriculum it diagnoses against.	basecamp-pilot.web.app

Running as a pilot at **basecamp-pilot.web.app**. A teacher photographs a pupil's worksheet, dictates an observation, or types a rubric entry. **Gemini multimodal analysis** reads the handwritten work and the audio reasoning, identifies the specific misconception behind a wrong answer rather than simply marking it wrong, and returns a diagnostic report with a targeted lesson plan and a parent SMS draft.

Offline is the default path, not a fallback. Assessments and voice clips are written to IndexedDB queues and reconciled by dedicated sync managers when an uplink returns, so a connection dropping mid-upload loses nothing and the teacher is never blocked waiting on a network.

Access is scoped by **organizationId** and **schoolId** across six roles, so a school network sees its own branches and nothing else. **Campus Gap Analysis** aggregates learning-gap prevalence across branches for leadership, and an **SEN inbox** routes screening alerts raised automatically during analysis to the responsible coordinator. Media handling is tuned for the low-specification tablets these classrooms already have, so adopting it costs a school nothing in new hardware.

- React 19
- Gemini Multimodal
- Firestore
- Cloud Functions
- IndexedDB
- Offline Sync
- RBAC

Production school information system, grading engine and report-card pipeline

ARCHIVE 5 yrs of records, 2021–2026	STACK React · Vite · Firebase · Gemini · WhatsApp Cloud API	LIVE sais-portal.web.app	USERS Teachers and leadership, primary and secondary
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The grading grid had to stop losing keystrokes. The master sheet replaced linear array scans with an `O(1)` hash-map lookup, isolated every cell behind `React.memo` with local state buffering, and added focus tracking so a background sync can never overwrite the cell a teacher is currently typing in.

Averages divide by subjects actually assessed

A pupil who transferred mid-year is not penalised for marks never taken. Two-decimal precision and deterministic competition ranking (1st, 2nd, 2nd, 4th) are enforced identically across analytics and printed reports.

VERIFIED IN SOURCE · GRADING ENGINE

Authorisation lives on the server. Term locks are enforced in Firestore security rules as well as the client, and the signed-in role resolves from a server-side whitelist — never from `localStorage`, which any user can edit. The client check is a convenience; the rules are the boundary.

Bulk PDF generation releases canvas GPU memory and unmounts React roots in a `finally` block, then packages a class into one ZIP with browser yielding. Parent delivery over the WhatsApp Cloud API records what actually happened via webhook rather than trusting a `200`, and inbound replies are stored for a human — a "STOP" is never acted on automatically, because withholding a child's report card would be the wrong outcome.

- React
- Firestore Rules
- Gemini API
- WhatsApp Cloud API
- JSZip
- RBAC

EduSchedulerPro

V1.0.0

Multi-constraint timetable, exam and duty-roster generator, sold as a one-time purchase

SOLVE TIME	STACK	VERIFICATION	STATUS
<3 min for 615 periods a week	Tauri v2 (Rust) · React 18 · Web Workers · Ed25519	Vitest · Playwright E2E · GitHub Actions CI	Released for delivery. Offline, no server, no telemetry.

Academic timetabling is a constraint satisfaction problem over teachers, rooms, class groups and curriculum splits. The solver is a hybrid: **dynamic minimum-remaining-values ordering with backtracking**, so it always assigns the most constrained slot next and unwinds cleanly when a branch dead-ends. It runs **inside a Web Worker**, so the interface stays responsive while it searches, and when it cannot place something it reports the unplaced lesson explicitly rather than producing a timetable that merely looks complete.

Divisions were the hard part. Schools running more than one bell schedule — a nursery section and a senior section on different day structures — break any scheduler comparing period indices, because period 3 is not the same hour in both. Conflict detection was rebuilt to compare *real clock times*, and that change had to propagate through the generator, exam rosters and duty assignment alike.

20 classes · 63 teachers · 615 periods · 0 unplaced

Verified against a real school export before the delivery build: no unplaced lessons, no curriculum gaps, no hard conflicts. The fixture is kept out of version control because it holds real staff data, so CI cannot run this gate — it is documented as a manual step before any release.

CHANGELOG.MD · V1.0.0, 2026-09-07

Multiple weeks of manual work became minutes of search, and it removed the residual room and staff clashes that had survived the manual process anyway. That ratio, not the raw speed, is the argument for solving this as a constraint problem.

Offline licensing with real cryptography. Activation verifies an Ed25519-signed licence key against a public key embedded in the binary. No account, no server call, and a tampered key is refused. Automatic restore points with undo/redo mean a destructive edit is recoverable without anyone having remembered to export a backup — and the restore is itself undoable. Exported problem reports carry **counts only, no staff, class, room or school names**, so a school can send one without sending its data.

- Rust
- Tauri v2
- MRV / Backtracking
- Web Workers
- Ed25519
- ExcelJS
- Playwright

Nexus Inventory

DELIVERED

Stock ledger, point of sale and direct thermal-receipt printing, built natively in Rust

TESTS 127 plus typecheck and cargo gates	STACK Rust · Tauri v2 · React 19 · rusqlite · aes-gcm · nusb	TARGETS Windows desktop shipped. Android signing outstanding.	STATUS Built to order for a named retail client in Ghana
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Thermal printing with no vendor driver. ESC/POS command bytes are assembled in Rust and sent two ways: over raw TCP to a network printer on port 9100 with a connect timeout, and over **USB bulk transfer via nusb**, with device enumeration and endpoint discovery done by hand. No print spooler, no third-party SDK, no driver install on the shop's machine.

Underneath is a native Rust IPC data layer over **rusqlite** in **WAL journal mode**, foreign keys enforced and a busy timeout set, carried through seven versioned schema migrations. Each is gated on **PRAGMA user_version** and wrapped in a transaction that rolls back on failure.

The SQL bridge is allow-listed

Queries crossing the IPC boundary are validated before execution. **DROP** is permitted only for the views migrations legitimately recreate, and **PRAGMA** is restricted to four known-safe pragmas. Anything else is refused rather than passed to SQLite.

SRC-TAURI/SRC/LIB.RS · READ IN SOURCE

Local backups are encrypted with **AES-256-GCM** using a per-backup random salt and a fresh 12-byte nonce, both prefixed to the ciphertext so a backup file is self-describing. Built on Tauri rather than Electron so the app uses the system webview instead of bundling a Chromium runtime — a deliberate trade for install size and memory headroom on the low-spec Windows machines this has to run on.

- Rust

rusqlite / WAL

ESC/POS

nusb (USB)

AES-256-GCM

TanStack

Victor — Ghanaian Damii Draughts Engine

JAN 2027

A full engine, training pipeline and client for a 10×10 variant that had no engine

SCALE	ENGINE	LEARNING	RELEASE
~30k lines of Rust, Python, TS	Bitboards · alpha-beta with PVS · Zobrist TT · quiescence	PyTorch NNUE · int16 quantisation · ≤5-piece tablebase	Engine complete. Network in training — launch January 2027.

The rules are the specification, and the tests pin them. Ghanaian Damii differs from international draughts in ways that change the search: mandatory capture but *free choice* — the maximum-capture rule is disabled, so a player may take any available path regardless of how many pieces it wins. Men capture backwards. Kings fly. There is no mid-capture promotion. A player loses immediately when their piece count drops below two, which cuts off long king chases. Each of the seven rules is named by the test that enforces it.

The board is **horizontally mirrored** relative to FMJD notation. I documented why that is a display convention rather than a symmetry: a true horizontal reflection flips square parity and maps dark squares onto light ones, so the real symmetry is a 180° rotation with colour swap. The mirror helper is marked display-only and **must never be used for data augmentation** — doing so would have silently poisoned the training set with illegal positions.

NNUE evaluation, trained end to end. Self-play data generation writes a dense 40-byte packed record per position, grouped by game id so splits happen per game rather than per position — consecutive plies are near-duplicates, and splitting naively would leak across train and validation. Weights quantise to int16 with quantisation-aware clamping, and every weight file carries a version header so an architecture mismatch is rejected rather than silently misread.

A published Elo claim, withdrawn in the repository

I had reported the network as roughly 100 Elo weaker than the hand-crafted evaluator. On re-examination the benchmark had been measuring the hand-crafted evaluator against *itself*, because the engine silently fell back when no trained weights were present. The figure is retracted in the README, the fallback is now unreachable from a search, and the benchmark exits rather than reporting a comparison it cannot make. Perft baselines are likewise labelled as not externally validated.

README.MD & ARCHITECTURE.MD · IN REPO

A **≤5-piece endgame tablebase** solved by retrograde analysis covers the 2v2 and 3v2 tiers — 18 million positions in 2v2 alone. The WASM bundle is size-tuned separately from the native profile at **337 KB**, and runs in a Web Worker so browser play never blocks the UI thread.

- Rust
- WebAssembly
- Bitboards
- Alpha-Beta / PVS
- NNUE
- PyTorch
- Tablebases

Full-stack e-commerce where the interesting problem is concurrency, not the shopfront

GUARDS	STACK	LIVE	STATUS
3 independent layers on stock	React 19 · Firestore · Cloud Functions · Recharts · Vitest	doxamix-foods.web.app	Storefront and admin dashboard for a Ghanaian food business

Catalogue, cart, checkout and order tracking for customers; sales analytics, inventory and order management for the owner. **The real problem is two customers buying the last bottle at the same moment.** A naive read-then-write oversells, and an oversold order in a food business means a refund and a lost customer.

Client guard → Firestore transaction → security rules

The cart checks availability so the customer gets fast feedback. A *Cloud Function* then re-reads every product inside a `runTransaction`, rejects the whole order if any line is short, and writes all deductions atomically — so the check and the write cannot be split by a competing order. Firestore rules permit stock writes to admins only. The client layer is convenience; the transaction is correctness; the rules are the boundary.

FUNCTIONS/SRC/INDEX.TS · READ IN SOURCE

A rejected order is **not silently dropped**. It is written back as `failed` with the reason that caused it — which product, stock available, quantity requested — so the owner can see what happened and tell the customer something true.

- React 19
- Firestore
- Cloud Functions
- Transactions
- RBAC
- Vitest

ADDITIONAL SYSTEMS

Academic Compliance

Cambridge lesson-plan auditing · live at `sais-lessons.web.app`

- Teachers upload a weekly plan as PDF or Word; Gemini audits it against **Cambridge lesson-planning standards** across ten subject departments, then bands it and routes it to the right head of department.
- Cited syllabus codes are checked against the class's actual stage — an out-of-stage citation caps the score — and a deterministic guard catches an unfilled template before it reaches the model.
- A **three-band compliance gate** clears at 76% and keeps the two bands below it distinct, with thresholds in one constants module the UI reads. **A threshold is never moved to improve the pass rate** — only if the rubric is wrong on the pedagogy, with the reason and decider recorded.

HecTech AI Report Generator

Google Apps Script · 39 class spreadsheets across three campuses

- A **Vault Library + Client Shell** architecture: engine and credentials in one private library, a thin shell bridge in each classroom spreadsheet. IT can revoke a single sheet from the authorisation registry without touching the others.
- **Auto-healing triggers** defeat the Apps Script six-minute ceiling — a long batch yields before timing out and schedules a trigger to resume itself.
- Hardened against the platform's failure modes: a cache-based **circuit breaker** on Gemini to protect the execution quota, formula-injection sanitising, a 30-minute OAuth token TTL, and retry-with-backoff on Drive PDF conversion.

QuantSports Engine

Quantitative odds research · places no bets, gives no advice

- Measures whether a pricing edge exists on two bookmakers, anchoring their lines against sharp consensus books and backtesting against historical price-and-result data.
- **The free-tier budget is enforced in code, not in discipline.** A credit ledger refuses any sweep that would breach a monthly or daily allowance before a request is sent.
- **Negative results are written down** — one bookmaker returning nothing usable on tier-one football is a recorded finding, not something quietly routed around.

3D Ludo Engine

Real-time 3D multiplayer board game

- A WebGL board in **Three.js** — marble slab, wooden framing, dynamic lighting, orbital camera with pinch-zoom and rotate that work on a phone.
- Three modes share one rules core: local pass-and-play, CPU opponents with distinct strategies, and **online multiplayer over WebSockets** with six-character room codes.
- Implements the **local variant** rather than the international rule set: a ransom mechanism holding captured pawns until redeemed, line kicks and reverse combo kicks, with a Vitest suite over move validation and win conditions.

METHOD — AGENT-ORCHESTRATED ENGINEERING

Agents implement; the constraints are mine

I build with **Claude Code**, **Antigravity** and **Cursor**. Each repository carries an **AGENTS.md** or **CLAUDE.md** setting out what to read first, what must pass before work is called done, and what must never be re-added without review.

Planning is version-controlled, not conversational

Structured **conductor/** tracks hold product guidelines, tech-stack decisions and workflow per project, so intent survives between sessions and tools. Agent skills are pinned by content hash in **skills-lock.json**.

Guardrails come from incidents

One of those files opens: *"most rules below exist because something went wrong once."* A rule forbidding a compliance threshold being moved to improve a pass rate. A provisioning script that refuses to run unfiltered. The written rule is guidance; the code guard is the backstop.

Generated code is assumed wrong until measured

Typecheck, unit tests, Playwright E2E and **cargo check** are gates an agent must clear before finishing. The retracted Elo figure and the "not externally validated" perft labels in this document exist because that discipline caught them.

HOW I WORK

Offline is a requirement, not a feature flag

Every product here assumes the network will fail. Write-behind queues in IndexedDB or SQLite, explicit reconciliation on reconnect, and no silent data loss when a sync races a user's keystrokes.

Verified at real scale, not demo scale

The scheduler's release gate is a real school export, held out of version control because it holds real staff data, and documented as a gap CI cannot cover.

The client check is never the security boundary

Term locks live in Firestore rules as well as the UI. Roles resolve server-side, never from browser storage. Stock deducts inside a transaction. An IPC SQL bridge allow-lists what may cross it.

Claims get retracted when they don't hold

I audit my own codebases in writing — including one audit that found plain-text credential storage in a project of mine and ranked it the single most serious finding. I would rather ship a documented unknown than an attractive number.

AVAILABILITY

OPEN TO ENGINEERING ROLES AND TECHNICAL CONSULTANCY

Strongest where the problem is genuinely hard — constraint solving, offline synchronisation, native desktop and hardware integration, search algorithms — and where correctness matters more than shipping speed. **Most systems above hold live school or customer data, so their repositories are private** — that is a condition of the work, not a gap in it. Four are running publicly at the addresses given. I can screen-share a walkthrough of any codebase here, talk through its architecture documents, or arrange repository access under NDA.